

Physics Publications with Major Personal Contributions

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May 10, 2026

1. A portrait of the Higgs boson by the CMS experiment ten years after the discovery.

Tumasyan, Armen et al.

[Nature 607 \(2022\) 60](#) | [arXiv:2207.00043](#) | [DOI:10.1038/s41586-022-04892-x](#) | 2022-06-30

2. Observation of the Production of Three Massive Gauge Bosons at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 125 \(2020\) 151802](#) | [arXiv:2006.11191](#) | [DOI:10.1103/PhysRevLett.125.151802](#) | 2020-06-19

3. Search for heavy Higgs bosons decaying to a top quark pair in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2020\) 171](#) | [arXiv:1908.01115](#) | [DOI:10.1007/JHEP04\(2020\)171](#) | 2019-08-02

4. Search for the production of $W^{\pm}W^{\pm}W^{\mp}$ events at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 100 \(2019\) 012004](#) | [arXiv:1905.04246](#) | [DOI:10.1103/PhysRevD.100.012004](#) | 2019-05-10

5. Search for dark matter produced in association with a Higgs boson decaying to a pair of bottom quarks in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 280](#) | [arXiv:1811.06562](#) | [DOI:10.1140/epjc/s10052-019-6730-7](#) | 2018-11-15

6. Search for dark matter in events with energetic, hadronically decaying top quarks and missing transverse momentum at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 027](#) | [arXiv:1801.08427](#) | [DOI:10.1007/JHEP06\(2018\)027](#) | 2018-01-25

7. Combined search for electroweak production of charginos and neutralinos in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[JHEP 03 \(2018\) 160](#) | [arXiv:1801.03957](#) | [DOI:10.1007/JHEP03\(2018\)160](#) | 2018-01-11

8. Search for standard model production of four top quarks with same-sign and multilepton final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 140](#) | [arXiv:1710.10614](#) | [DOI:10.1140/epjc/s10052-018-5607-5](#) | 2017-10-29

9. Search for new phenomena in final states with two opposite-charge, same-flavor leptons, jets, and missing transverse momentum in pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2018\) 076](#) | [arXiv:1709.08908](#) | [DOI:10.1007/JHEP03\(2018\)076](#) | 2017-09-26

10. Search for electroweak production of charginos and neutralinos in WH events in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 11 \(2017\) 029](#) | [arXiv:1706.09933](#) | [DOI:10.1007/JHEP11\(2017\)029](#) | 2017-06-29

11. Search for top squark pair production in pp collisions at $\sqrt{s}=13$ TeV using single lepton events

Sirunyan, Albert M et al.

[JHEP 10 \(2017\) 019](#) | [arXiv:1706.04402](#) | [DOI:10.1007/JHEP10\(2017\)019](#) | 2017-06-14

12. Search for new phenomena with the M_{T2} variable in the all-hadronic final state produced in proton–proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 710](#) | [arXiv:1705.04650](#) | [DOI:10.1140/epjc/s10052-017-5267-x](#) | 2017-05-12

13. Search for physics beyond the standard model in events with two leptons of same sign, missing transverse momentum, and jets in proton–proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 578](#) | [arXiv:1704.07323](#) | [DOI:10.1140/epjc/s10052-017-5079-z](#) | 2017-04-24

14. Searches for pair production of third-generation squarks in $\sqrt{s}=13$ TeV pp collisions

Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 327](#) | [arXiv:1612.03877](#) | [DOI:10.1140/epjc/s10052-017-4853-2](#) | 2016-12-12

15. Search for new physics in final states with two opposite-sign, same-flavor leptons, jets, and missing transverse momentum in pp collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[JHEP 12 \(2016\) 013](#) | [arXiv:1607.00915](#) | [DOI:10.1007/JHEP12\(2016\)013](#) | 2016-07-04

16. Search for new physics in same-sign dilepton events in proton–proton collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 439](#) | [arXiv:1605.03171](#) | [DOI:10.1140/epjc/s10052-016-4261-z](#) | 2016-05-10

17. Measurements of $t\bar{t}$ charge asymmetry using dilepton final states in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 760 \(2016\) 365](#) | [arXiv:1603.06221](#) | [DOI:10.1016/j.physletb.2016.07.006](#) | 2016-03-20

18. Search for new physics with the M_{T2} variable in all-jets final states produced in pp collisions at $\sqrt{s}=13$ TeV

Khachatryan, Vardan et al.

[JHEP 10 \(2016\) 006](#) | [arXiv:1603.04053](#) | [DOI:10.1007/JHEP10\(2016\)006](#) | 2016-03-13

19. Measurements of $t\bar{t}$ spin correlations and top quark polarization using dilepton final states in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, V. et al.

[Phys.Rev.D 93 \(2016\) 052007](#) | [arXiv:1601.01107](#) | [DOI:10.1103/PhysRevD.93.052007](#) | 2016-01-06

20. Measurement of the W^+W^- cross section in pp collisions at $\sqrt{s} = 8$ TeV and limits on anomalous gauge couplings

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 401](#) | [arXiv:1507.03268](#) | [DOI:10.1140/epjc/s10052-016-4219-1](#) | 2015-07-12

21. Search for Physics Beyond the Standard Model in Events with Two Leptons, Jets, and Missing Transverse Momentum in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 04 \(2015\) 124](#) | [arXiv:1502.06031](#) | [DOI:10.1007/JHEP04\(2015\)124](#) | 2015-02-20

22. Observation of the rare $B^0_s \rightarrow \mu^+ \mu^-$ decay from the combined analysis of CMS and LHCb data

Khachatryan, Vardan et al.

[Nature 522 \(2015\) 68](#) | [arXiv:1411.4413](#) | [DOI:10.1038/nature14474](#) | 2014-11-17

23. Study of vector boson scattering and search for new physics in events with two same-sign leptons and two jets

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 114 \(2015\) 051801](#) | [arXiv:1410.6315](#) | [DOI:10.1103/PhysRevLett.114.051801](#) | 2014-10-23

24. Searches for electroweak neutralino and chargino production in channels with Higgs, Z, and W bosons in pp collisions at 8 TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 90 \(2014\) 092007](#) | [arXiv:1409.3168](#) | [DOI:10.1103/PhysRevD.90.092007](#) | 2014-09-10

25. Measurement of Top Quark-Antiquark Pair Production in Association with a W or Z Boson in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 74 \(2014\) 3060](#) | [arXiv:1406.7830](#) | [DOI:10.1140/epjc/s10052-014-3060-7](#) | 2014-06-30

26. Searches for Electroweak Production of Charginos, Neutralinos, and Stopped Decaying to Leptons and W, Z, and Higgs Bosons in pp Collisions at 8 TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 74 \(2014\) 3036](#) | [arXiv:1405.7570](#) | [DOI:10.1140/epjc/s10052-014-3036-7](#) | 2014-05-29

27. Measurements of the $t\bar{t}$ Charge Asymmetry Using the Dilepton Decay Channel in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 04 \(2014\) 191](#) | [arXiv:1402.3803](#) | [DOI:10.1007/JHEP04\(2014\)191](#) | 2014-02-16

28. Measurement of Higgs Boson Production and Properties in the WW Decay Channel with Leptonic Final States

Chatrchyan, Serguei et al.

[JHEP 01 \(2014\) 096](#) | [arXiv:1312.1129](#) | [DOI:10.1007/JHEP01\(2014\)096](#) | 2013-12-04

29. Search for New Physics in Events with Same-Sign Dileptons and Jets in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[JHEP 01 \(2014\) 163](#) | [arXiv:1311.6736](#) | [DOI:10.1007/JHEP01\(2014\)163](#) | 2013-11-26

30. Measurements of $t\bar{t}$ Spin Correlations and Top-Quark Polarization Using Dilepton Final States in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 112 \(2014\) 182001](#) | [arXiv:1311.3924](#) | [DOI:10.1103/PhysRevLett.112.182001](#) | 2013-11-15

31. Search for Top-Squark Pair Production in the Single-Lepton Final State in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2677](#) | [arXiv:1308.1586](#) | [DOI:10.1140/epjc/s10052-013-2677-2](#) | 2013-08-07

32. Measurement of the W^+W^- Cross Section in pp Collisions at $\sqrt{s} = 7$ TeV and Limits on Anomalous $WW\gamma$ and WWZ Couplings

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2610](#) | [arXiv:1306.1126](#) | [DOI:10.1140/epjc/s10052-013-2610-8](#) | 2013-06-05

33. Measurement of W^+W^- and ZZ Production Cross Sections in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 721 \(2013\) 190](#) | [arXiv:1301.4698](#) | [DOI:10.1016/j.physletb.2013.03.027](#) | 2013-01-20

34. Search for New Physics in Events with Same-Sign Dileptons and b Jets in pp Collisions at $\sqrt{s}=8$ TeV

Chatrchyan, Serguei et al.

[JHEP 03 \(2013\) 037](#) | [arXiv:1212.6194](#) | [DOI:10.1007/JHEP03\(2013\)037](#) | 2012-12

35. Search for Electroweak Production of Charginos and Neutralinos using Leptonic Final States in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 11 \(2012\) 147](#) | [arXiv:1209.6620](#) | [DOI:10.1007/JHEP11\(2012\)147](#) | 2012-09

36. Observation of a New Boson at a Mass of 125 GeV with the CMS Experiment at the LHC

Chatrchyan, Serguei et al.

[Phys.Lett.B 716 \(2012\) 30](#) | [arXiv:1207.7235](#) | [DOI:10.1016/j.physletb.2012.08.021](#) | 2012-07

37. Search for New Physics in Events with Opposite-Sign Leptons, Jets, and Missing Transverse Energy in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 718 \(2013\) 815](#) | [arXiv:1206.3949](#) | [DOI:10.1016/j.physletb.2012.11.036](#) | 2012-06

38. Search for New Physics with Same-Sign Isolated Dilepton Events with Jets and Missing Transverse Energy

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 109 \(2012\) 071803](#) | [arXiv:1205.6615](#) | [DOI:10.1103/PhysRevLett.109.071803](#) | 2012-05

39. Search for New Physics in Events with Same-Sign Dileptons and b-Tagged Jets in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 08 \(2012\) 110](#) | [arXiv:1205.3933](#) | [DOI:10.1007/JHEP08\(2012\)110](#) | 2012-05

40. Search for Physics Beyond the Standard Model in Events with a Z Boson, Jets, and Missing Transverse Energy in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 716 \(2012\) 260](#) | [arXiv:1204.3774](#) | [DOI:10.1016/j.physletb.2012.08.026](#) | 2012-04

41. Search for Heavy, Top-Like Quark Pair Production in the Dilepton Final State in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 716 \(2012\) 103](#) | [arXiv:1203.5410](#) | [DOI:10.1016/j.physletb.2012.07.059](#) | 2012-03

42. Search for the Standard Model Higgs Boson in the $H \rightarrow ZZ \rightarrow 2\ell 2\nu$ Channel in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 03 \(2012\) 040](#) | [arXiv:1202.3478](#) | [DOI:10.1007/JHEP03\(2012\)040](#) | 2012-02

43. Search for the Standard Model Higgs Boson Decaying to W^+W^- in the Fully Leptonic Final State in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 710 \(2012\) 91](#) | [arXiv:1202.1489](#) | [DOI:10.1016/j.physletb.2012.02.076](#) | 2012-02

44. Search for Same-Sign Top-Quark Pair Production at $\sqrt{s}=7$ TeV and Limits on Flavour Changing Neutral Currents in the Top Sector

Chatrchyan, Serguei et al.

[JHEP 08 \(2011\) 005](#) | [arXiv:1106.2142](#) | [DOI:10.1007/JHEP08\(2011\)005](#) | 2011-06

45. Measurement of the $t\bar{t}$ production cross section and the top quark mass in the dilepton channel in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 07 \(2011\) 049](#) | [arXiv:1105.5661](#) | [DOI:10.1007/JHEP07\(2011\)049](#) | 2011-05

46. Search for new physics with same-sign isolated dilepton events with jets and missing transverse energy at the LHC

Chatrchyan, Serguei et al.

[JHEP 06 \(2011\) 077](#) | [arXiv:1104.3168](#) | [DOI:10.1007/JHEP06\(2011\)077](#) | 2011-04

47. Search for Physics Beyond the Standard Model in Opposite-Sign Dilepton Events at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2011\) 026](#) | [arXiv:1103.1348](#) | [DOI:10.1007/JHEP06\(2011\)026](#) | 2011-03

48. First Measurement of the Cross Section for Top-Quark Pair Production in Proton-Proton Collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 695 \(2011\) 424](#) | [arXiv:1010.5994](#) | [DOI:10.1016/j.physletb.2010.11.058](#) | 2010-10

49. Measurement of beauty production from dimuon events at HERA

Chekanov, S. et al.

[JHEP 02 \(2009\) 032](#) | [arXiv:0811.0894](#) | [DOI:10.1088/1126-6708/2009/02/032](#) | 2008-09

50. Measurement of Beauty Quark Cross Sections in Photoproduction with the ZEUS Experiment at the Electron Proton Collider HERA

Gutsche, Oliver et al.

[DOI:10.3204/DESY-THESIS-2005-010](#) | 2005-03

51. Development of the Trigger Algorithm for the MONOLITH Experiment

Gutsche, Oliver et al.

2001-05